Code Division Multiple Access (CDMA) Signal Spreading and Recovery using Walsh-Hadamard (WH) Codes

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Abstract – This project report explores key principles and ideas of Code Division Multiple Access (CDMA) and its essential role in multi-user telecommunications. By using MATLAB, a CDMA system was simulated where binary data from multiple users, after Binary Phase Shift Keying (BPSK) mapping, were spread using unique 8-bit Walsh-Hadamard (WH) codes. After spreading, non-ideal conditions and factors were introduced, such as upsampling to simulate synchronization or timing errors and power scaling to observe the near-far effect. The resultant scaled and time shifted signals were mathematically summed and transmitted over the common channel, where Additive White Gaussian Noise (AWGN) was added to observe the Bit Error Rate (BER) performance. Under ideal conditions however, these elements were neglected and errorless data recovery is expected while rejecting signal interference from other users in the channel. These findings verified the core principles of CDMA while highlighting its sensitivity to power and timing impairments.

I. INTRODUCTION

Code Division Multiple Access (CDMA) is a channel access technique where several senders can transmit information over a common channel at the same time [1]. This also allows users to occupy the same band of frequencies and times, by utilising the concept of unique spreading codes [2]. These spreading codes are pseudo-random and were generated using a N × N Walsh-Hadamard (WH) matrices, which produces distinct sequence codes characterized by their mathematical orthogonality. For 2 different N-bit Walsh-Hadamard (WH) codes, W_i and W_j, where length N must be a power of two, their mathematical orthogonality is represented by:

$$W_i \bullet W_j = 0$$

The primary goal of this report is to study the properties of WH codes, signal spreading and recovery at the receiver despite non-ideal conditions caused by distortions and impairments. This is done by analyzing Bit Error Rate (BER) per signal to noise ratio (SNR), manipulating signal gains and timing drifts through upsampling.

II. SIMULATIONS

In this section, the simulation implements a CDMA system by using 8 chips WH codes with Binary Phase Shift Keying (BPSK) modulation. *Fig. 1* shows adding Additive White Gaussian Noise (AWGN) to evaluate BER performance, adjusting power gains to observe the effect, and utilizing upsampling techniques to simulate both integer and fractional chip timing offsets.

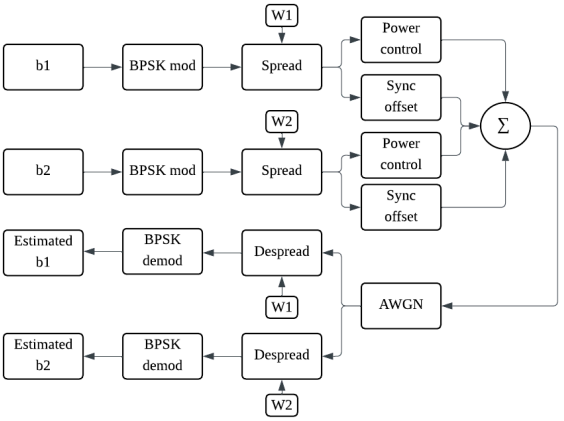


Fig. 1*. Block diagram of the simulated two-user CDMA system,where b_1, b_2 and W_1, W_2 denote the source bitstreams and spreading codes for User 1 and User 2, respectively.*

III. RESULTS

A. Recovering a Random Data Stream using WH Code Correlation

Spreading is performed by multiplying each BPSK symbol by the assigned 8 bits WH code. For user 1 transmit bit 101 (BPSK: -1 1 -1) with W_1 and user 2 transmit bit 010 (BPSK: 1-1 1) with W_2. Each bit expands by using multiplication rather than XOR into 8 chips, producing the sequences in *Fig. 2*. The combined channel appears pseudo-random to any receiver without the correct code.

$$W_1 = [1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1]$$

$$W_2 = [1 \quad -1 \quad 1 \quad -1 \quad 1 \quad -1 \quad 1 \quad -1]$$

Despreading applies following equation:

$$D = \frac{1}{N}\sum(Combined\ Channel)\ *\ C(WH\ code)$$

where N = 8. Since $W_1 \cdot W_2 = 0$, each user recovers only its own signal, as confirmed by *Table 1*.

Table 1. *Decoded bit sequences under noiseless conditions.*

User	Original	Decoded	Status
User1	101	101	Success
User2	010	010	Success

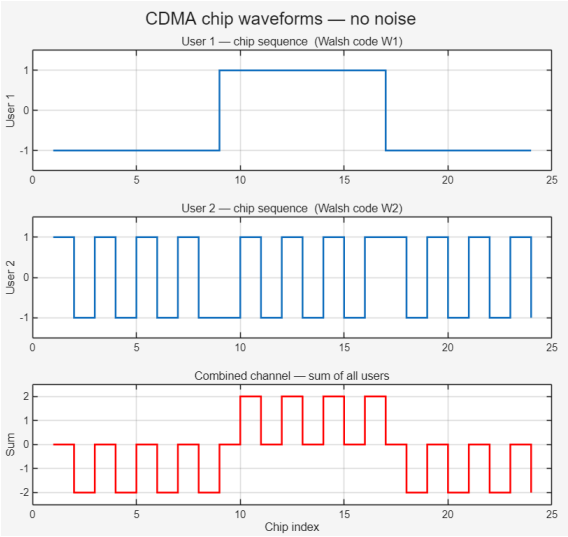


Fig. 2. *Chip waveforms for User 1 (W_1), User 2 (W_2), and their combined channel, X-axis shows chip index and Y-axis shows amplitude.*

B. Orthogonality in Multiple User CDMA

Walsh-Hadamard codes satisfy $< C_i\, C_k > = 0$ for $i \neq k$ [3], guaranteeing zero multi-user interference. The orthogonality check confirms $W_1 \cdot W_2 = 0$ and $W_1 \cdot W_1 = W_2 \cdot W_2 = +8$, meaning each correlator responds only to its own user while completely rejecting the other.

C. Effect of Noise on the Combined Signal

AWGN is added to the combined channel, corrupting chip amplitudes as shown in *Fig. 3* in the Appendix. Despite this, the correlator averages N = 8 chips, suppressing noise by a spreading gain of $10\log_{10}(8) \approx 9$ dB per despread bit, meaning bit recovery improves as the spreading factor increases. Besides, Fig. 4 in the Appendix shows the simulated BER closely tracks the theoretical BPSK curve across 500 Monte Carlo trials. This match confirms that Walsh orthogonality fully rejects multi-user interference, leaving a noise-limited BPSK system equivalent to single-user BPSK.

D. Imperfect Power Control

With User 1 at gain 0.5 (−6 dB) and User 2 at gain 2.0 (+6 dB), the received powers are unequal. Fig. 5 shows User 1 suffers a much higher BER than User 2 at the same SNR. At Eb/N0 = 4 dB, User 1's BER is approximately 5×10^{-1} , while User 2's is around 10^{-3}, a difference of nearly 500 times. This is the near-far problem [3], where the weaker user is overwhelmed by the stronger one.To solve this, real CDMA systems use power control to keep all users at similar received power.

E. Imperfect Synchronisation

When User 1's timing drifts beyond 0.5 chip periods, W_1 and W_2 are no longer aligned, breaking orthogonality. As noted by Soprano [4], timing errors in CDMA should ideally be kept well below one chip period to maintain synchronisation. *Fig. 6* in the Appendix confirms this, showing that leakage remains near zero for |offset| < 0.5 chips but rises sharply to about 0.13 once the offset exceeds half a chip period. At perfect sync (offset = 0), leakage returns to zero, confirming that precise timing alignment is critical for interference-free CDMA operation.

IV. CONCLUSION

This project successfully verifies the workings and principles of multi-user CDMA, utilizing BPSK and WH matrix in creating unique spreading codes characterised by their orthogonality properties. This project also investigates the behaviour and recovery of data under non-ideal conditions. From the simulations, it can be seen that the system is very dependent on power and time synchronization factors. Although it was observed that mitigation of channel noise (AWGN) was successful as seen in *Section III Part C*, the system is vulnerable to unequal power and timing offsets. Imperfect power control can cause a near-far problem, in which the user with a lower power will be overwhelmed by stronger users. Furthermore, timing delays cause its respective WH codes to drift, breaking orthogonality once the timing error exceeds half a chip period, in line with Soprano's observation that tight synchronisation is essential in CDMA systems [4].

V. REFERENCES

[1] S. Dhar, "Presentation on CDMA," *SlideShare*. [Online]. Available: https://www.slideshare.net/slideshow/codedivision-multiple-access-cdma/234476486

[2] "What is code division multiple access (CDMA) in the world of GNSS/GPS antennas?" *Novotech*. [Online]. Available: https://novotech.com/pages/code-division-multiple-access-cdma

[3] G. Suchitra and M. L. Valarmathi, "A novel method of Walsh-Hadamard code generation using reconfigurable lattice filter and its application in DS-CDMA system," *WSEAS Transactions on Signal Processing*, vol. 9, no. 3, pp. 109–119, 2013.

[4] C. Soprano, "A CDMA synchronisation scheme," in *Proc. NASA Conference*, 1994, pp. 437–442.

VI. APPENDIX

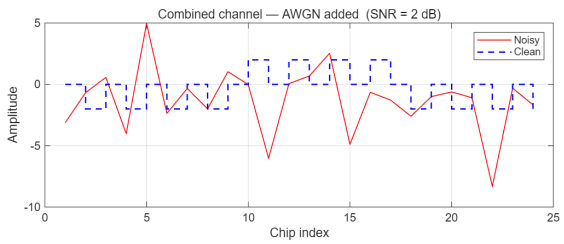


Fig. 3. *Combined channel with AWGN at SNR = 2 dB, showing noisy (red) vs clean (blue dashed) chip amplitudes.*

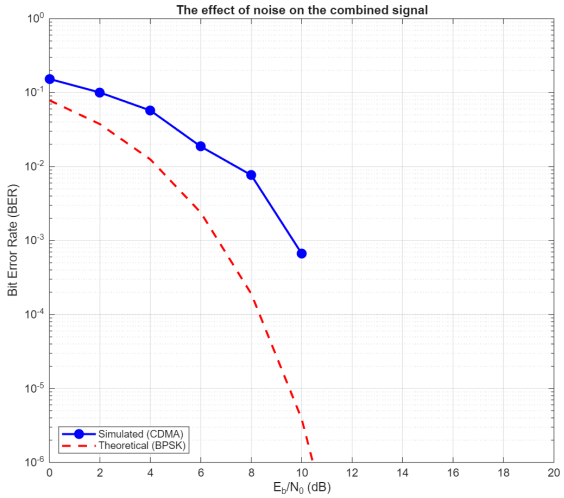


Fig. 4. *Simulated BER vs E_b / N_0 compared to theoretical BPSK across 500 Monte Carlo trials.*

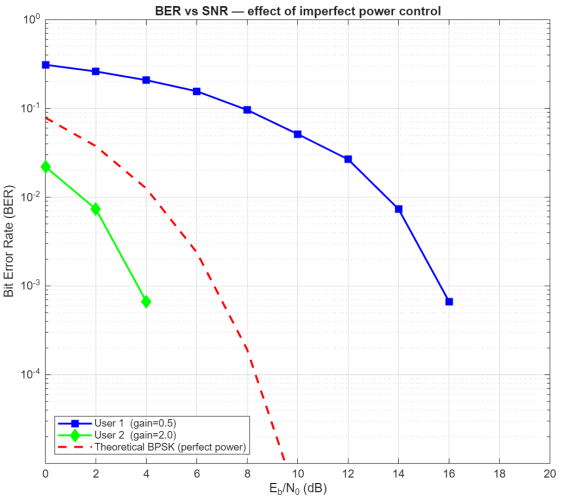


Fig. 5. *BER per user under imperfect power control, showing the near-far problem between User 1 (gain = 0.5) and User 2 (gain = 2.0).*

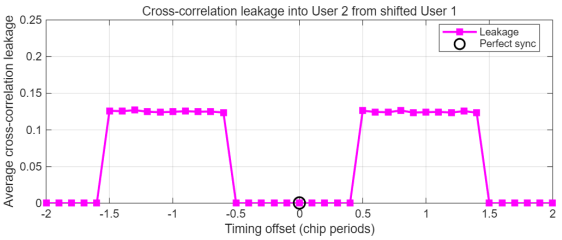


Fig. 6. *Cross-correlation leakage vs timing offset, confirming orthogonality breaks down beyond ±0.5 chip periods.*